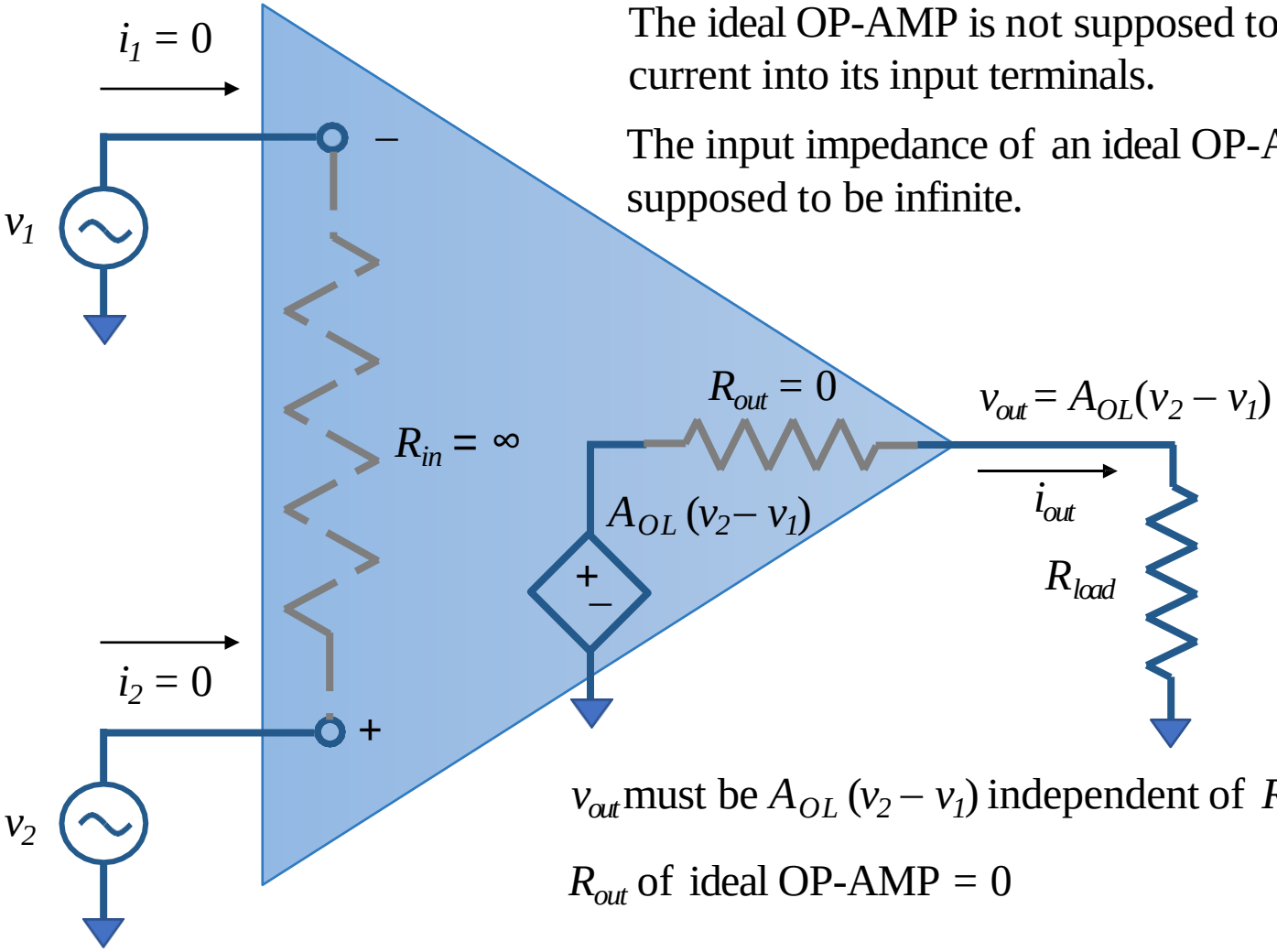
Analog Electronics II: Op-amp & 555 Timer

Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

Lecture # 4

Ideal OP-AMP equivalent circuit



The ideal OP-AMP is not supposed to draw any current into its input terminals.

The input impedance of an ideal OP-AMP is supposed to be infinite.

v_{out} must be $A_{OL}\,(v_2 - v_1)$ independent of R_{load} or i_{out}.

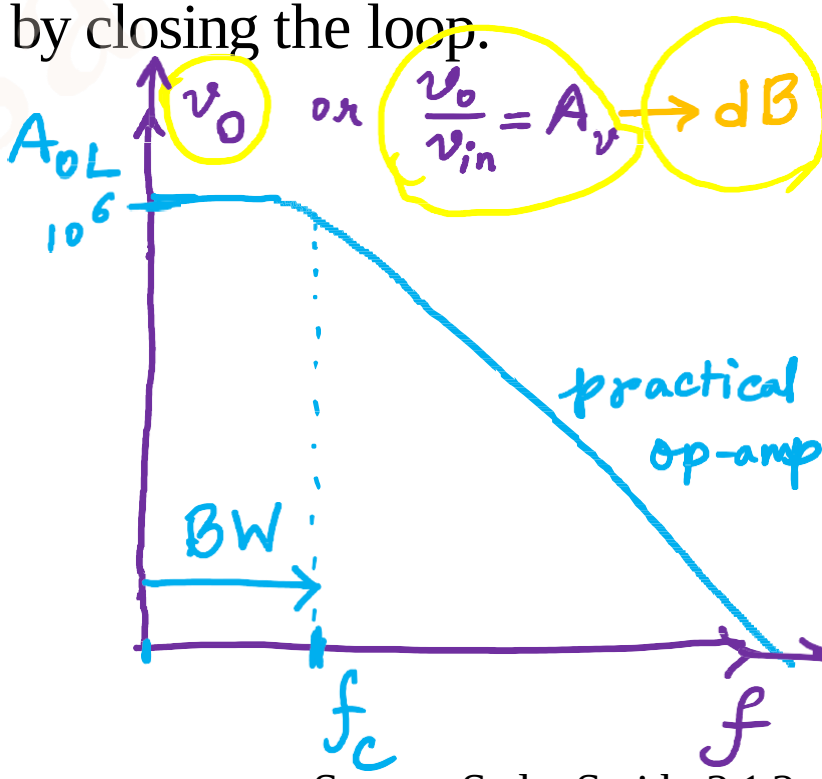
R_{out} of ideal OP-AMP = 0

Source: Sedra Smith, 2.1.2

Ideal OP-AMP: gain and bandwidth

- ✓ Infinite bandwidth: Gain remains constant down to zero frequency and up to infinite frequency.
- ✓ Infinite open-loop gain. But in most circuit designs, OP-AMP is not used in the open-loop configuration. OP-AMP is usually used with other circuit elements to apply feedback by closing the loop.

A_{OL}	10^{5-6} (practical)	∞ (ideal)
BW	Finite	∞



Source: Sedra Smith, 2.1.2

©Dr. Nafis Alam

Characteristic	Ideal Op-Amp	Practical Op-Amp
Input Impedance (Z_{in})	∞ (Infinite)	High (e.g., MΩ to GΩ range)
Output Impedance (Z_{out})	0 (Zero)	Low (e.g., tens to hundreds of ohms)
Open-Loop Gain (A_{OL})	∞ (Infinite)	High but finite (e.g., 10^5 to 10^6)
Common-Mode Gain (A_{CM})	0 (Zero)	Small but nonzero
Common-Mode Rejection Ratio (CMRR)	∞ (Infinite)	High but finite (e.g., 80–120 dB)
Bandwidth	∞ (Infinite)	Limited (depends on Gain-Bandwidth Product)
Slew Rate	∞ (Infinite)	Limited (e.g., V/µs range)
Input Offset Voltage	0 V	Small (e.g., mV to µV range)
Input Bias Current	0 A	Small (nA to pA range)
Power Supply Requirement	None (assumed ideal)	Finite voltage range (e.g., ±15V, 0-5V)
Output Voltage Swing	Can rea pply rails	Limited (e.g., close but not equal to supply rails)

Ideal vs Practical OP-AMP: characteristics

Source: Sedra Smith, 2.1.2

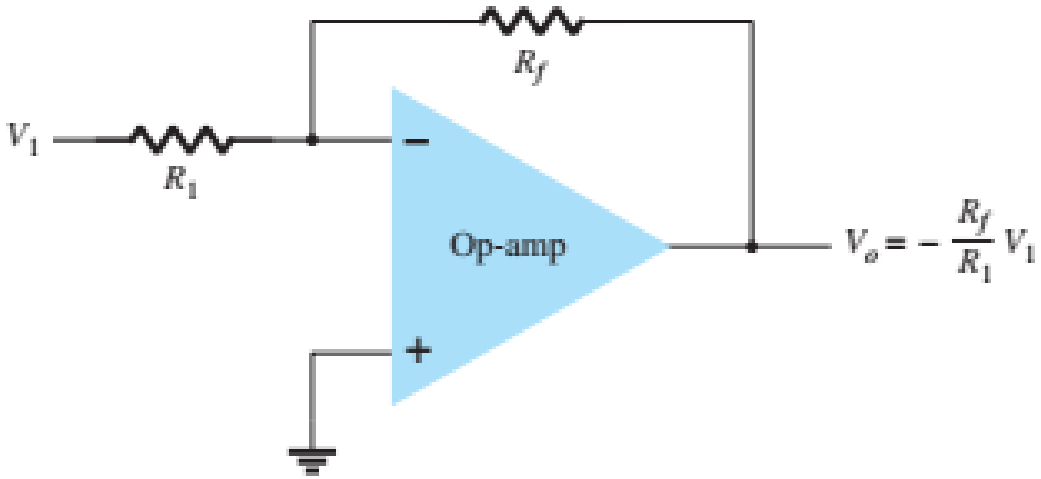
Inverting Amplifier

1. ***Negative feedback:*** between the output and the ***inverting*** input terminals. Due to the feedback, this is a closed-loop configuaration.
2. ***Inverting configuration:*** Signal is connected at the inverting input terminal. So, there will be phase inversion of signal at the output.

$$V_o = -\frac{R_f}{R_1} V_1$$

the voltage gain

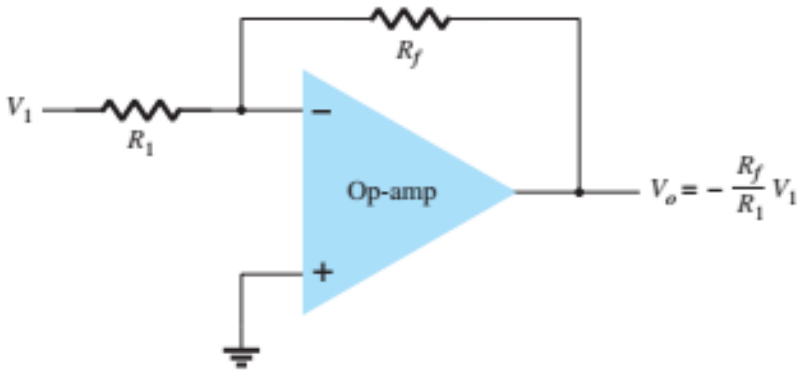
$$A_v = -\frac{R_f}{R_i}$$



Inverting Amplifier

The input impedance is

$$Z_{in} = R_1$$



Constant-Magnitude Gain

If R_f is some multiple of R_1, the overall amplifier gain is a constant. For example, if $R_f = 10R_1$, then

$$\text{Voltage gain} = -\frac{R_f}{R_1} = -10$$

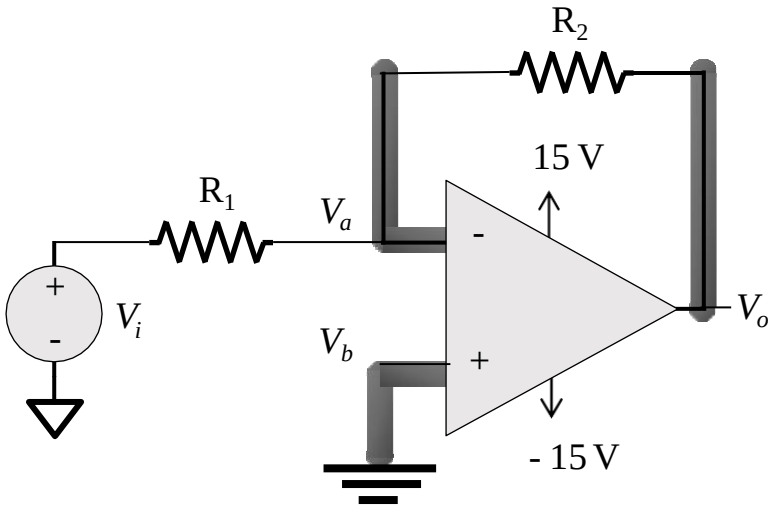
Unity Gain

If $R_f = R_1$, the gain is

$$\text{Voltage gain} = -\frac{R_f}{R_1} = -1$$

Virtual short circuit & virtual ground

1. ***Virtual short circuit*** (not physical): between the non-inverting and inverting input terminals
   - ✓ The two input terminals track each other in potential
2. ***Virtual ground*** (not physical): at the inverting input terminal

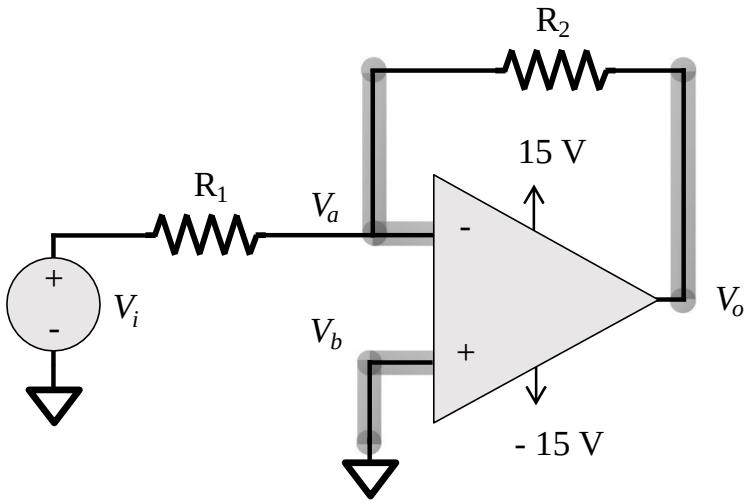


Source: Sedra Smith 2.2

Inverting amplifier: closed-loop gain

1. Open loop gain A_{OL} varies from one op-amp to the other $v_o \quad A_{OL} \cdot v_d$
2. Closed loop gain can be very accurate, stable, and predictable (trading gain for accuracy)

Closed-loop gain, $A_{CL} = \frac{v_0}{v_I}$ $v_0 = A_{CL} \cdot v_i$



Inverting amplifier: closed-loop gain

KCL @ node a, $I_1 = I_2 + 0A \Rightarrow I_1 = I_2$

(R_1 & R_2 are in series)

$$\frac{v_o}{v_I} = -\frac{R_2}{R_1}$$

inversion

$$\Rightarrow \frac{V_i - 0V}{R_1} \qquad \frac{0V - V_o}{R_2}$$

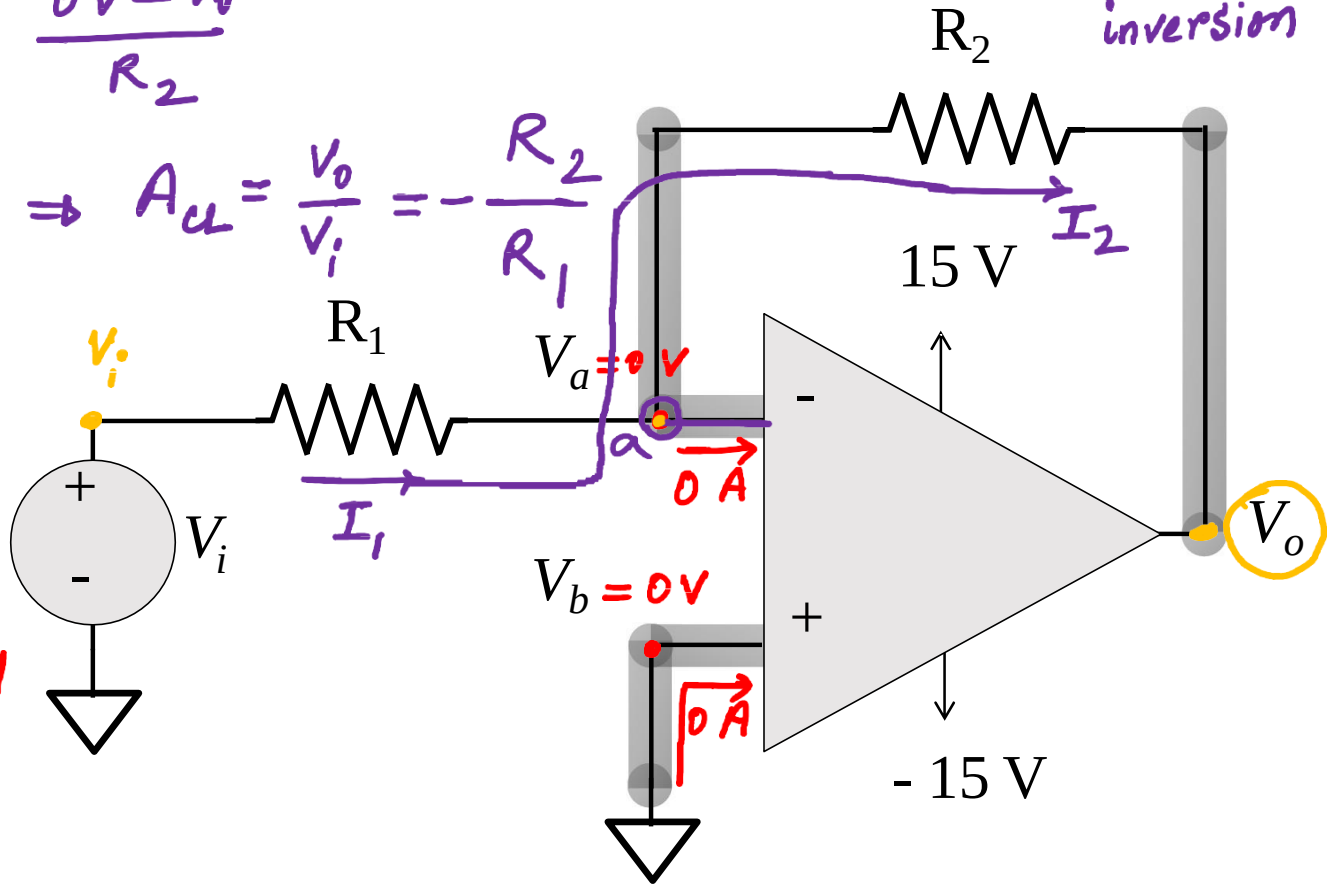
$$\Rightarrow \frac{V_i}{R_1} = -\frac{V_o}{R_2} \Rightarrow A_{CL} = \frac{V_o}{V_i} = -\frac{R_2}{R_1}$$

CL virtual S.C.

$V_a = V_b$

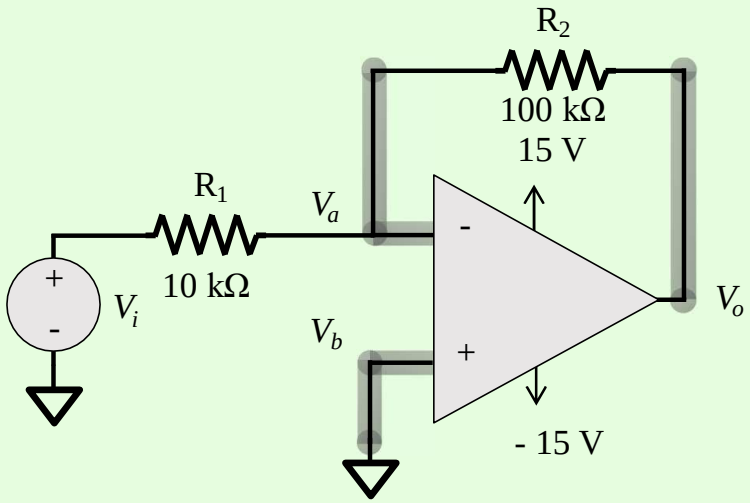
inv. amp.: virtual gnd

$V_a = 0V = V_b$



Inverting amplifier gain $A_{CL} = -\frac{R_2}{R_1}$

(a)



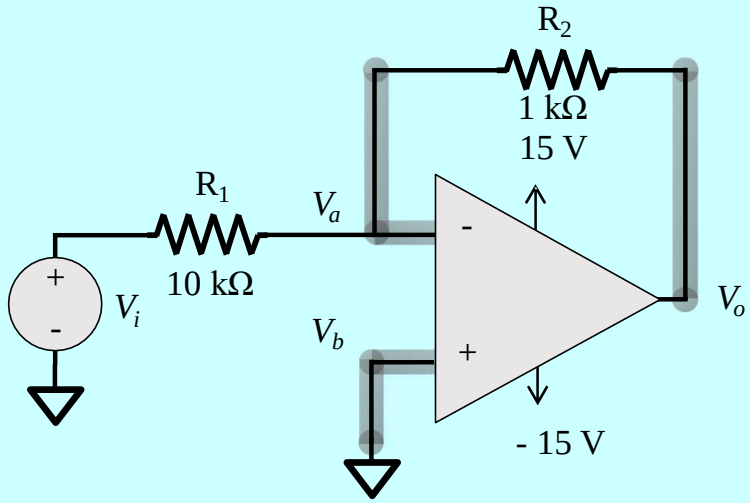
Amplification when $R_2 > R_1$

$$A_{CL} = -\frac{R_2}{R_1} = -\frac{100\ k\Omega}{10\ k\Omega} = -10\ \frac{V}{V}$$

$$\Rightarrow |A_{CL}| = 10\ \frac{V}{V} > 1$$

$$\Rightarrow \left|\frac{v_o}{v_i}\right| > 1 \Rightarrow |v_o| > |v_i|$$

(b)



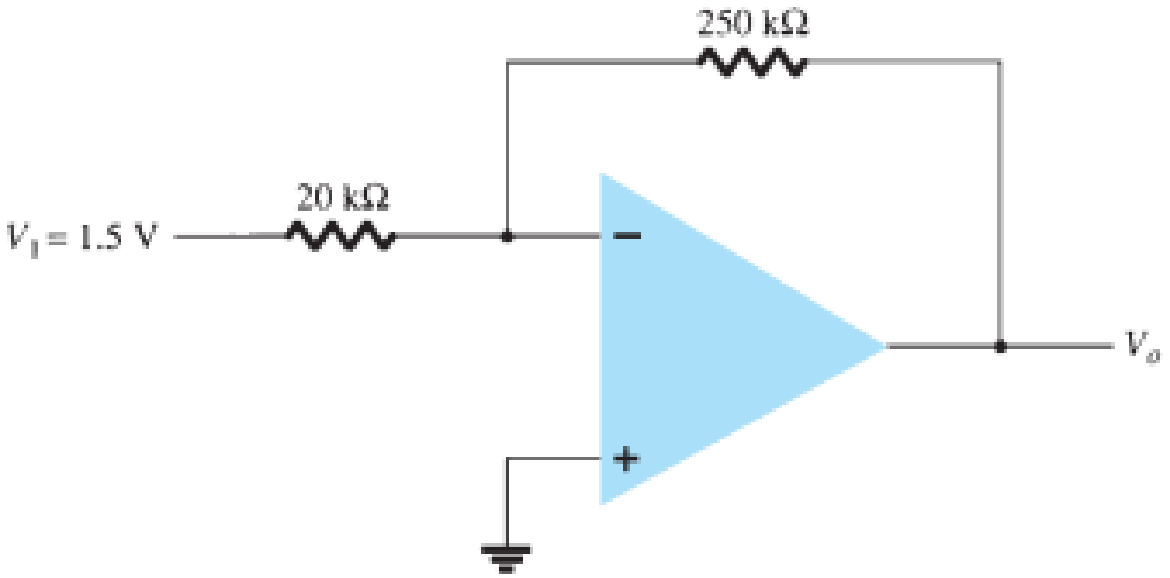
Attenuation when $R_2 < R_1$

$$A_{CL} = -0.1\ \frac{V}{V} \Rightarrow |A_{CL}| = 0.1\ \frac{V}{V} < 1$$

$$\Rightarrow \left|\frac{v_o}{v_i}\right| < 1 \Rightarrow |v_o| < |v_i|$$

Practice Problem

What is the output voltage in the circuit?



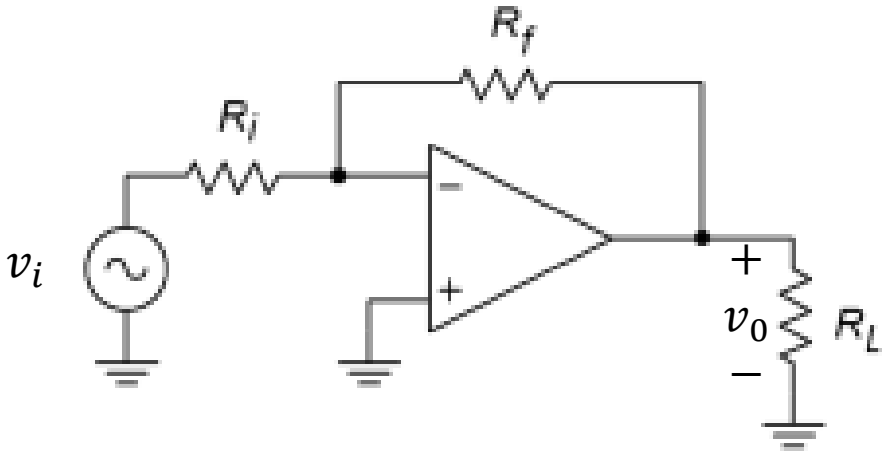
$$V_o = -\frac{R_f}{R_1} V_1$$

Practice Problem

Design an inverting amplifier with a gain of 10 and an input impedance of 15 kΩ.

The input impedance tells us what R_i must be

$$Z_{in} = R_i$$
$$R_i = 15\,\text{k}$$



Knowing R_i, solve for R_f:

$$A_v = -\frac{R_f}{R_i}$$
$$R_f = R_i(-A_v)$$
$$R_f = 15\,\text{k} \times (-(-10))$$
$$R_f = 150\,\text{k}$$

Practice Problem

What is the system input impedance ? What is the system gain?

